Engine Model(s): PT6A

72-30-02

REAR FIRESEAL MOUNT RING - MAINTENANCE PRACTICES

1. General

A. It is not a necessary practice to remove the rear fireseal mount ring during maintenance of the engine. Removal should be dictated by the extent of repair necessary as determined during inspection.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC03-001	Oil, Engine Lubricating
PWC05-061	Cloth, Abrasive
PWC08-002	Adhesive, Fluorosilicone
PWC11-014	Alcohol, Isopropyl
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Removal/Installation

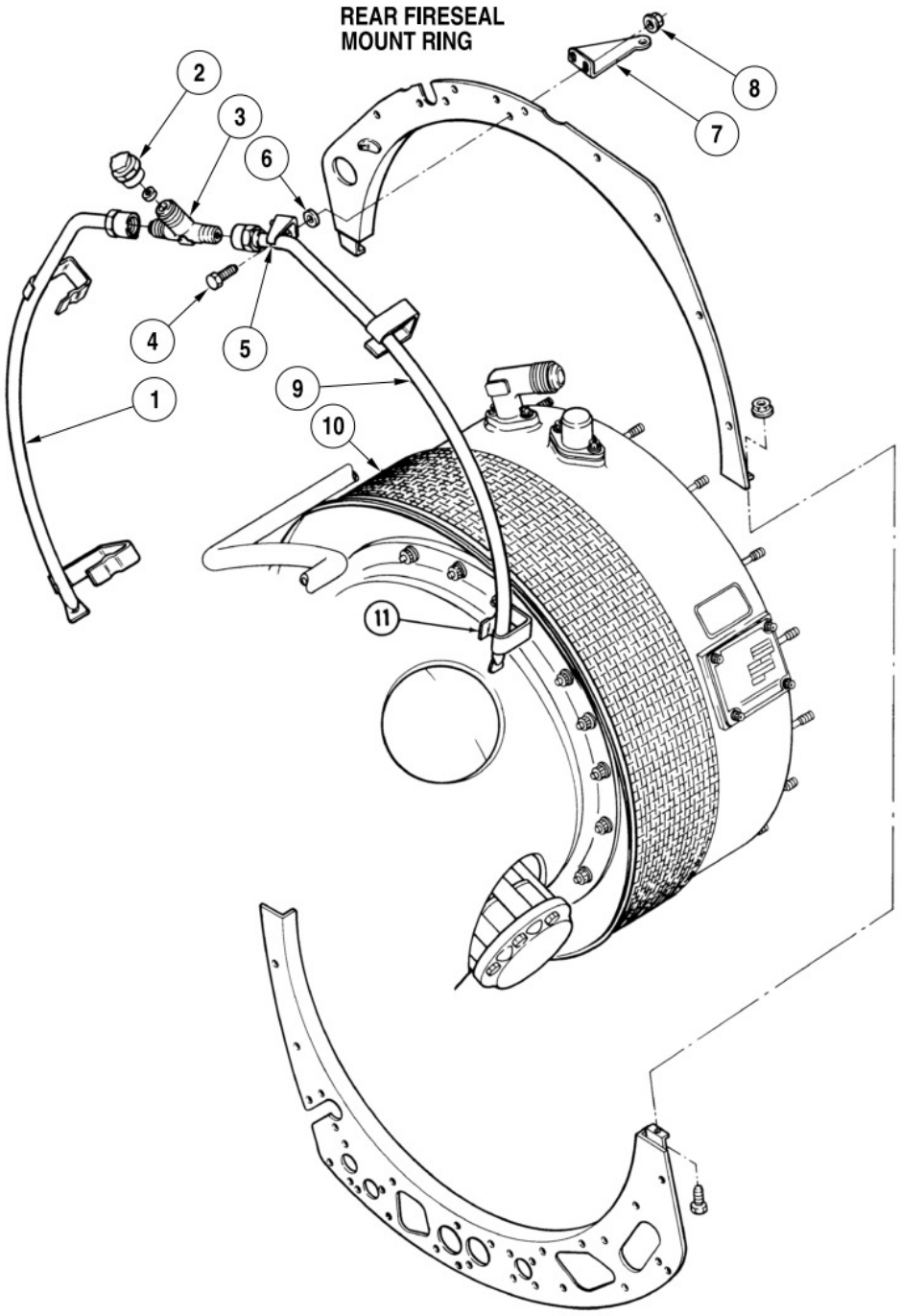
It is not a necessary practice to remove the rear fireseal mount ring at maintenance of the engine. Removal should be dictated by the extent of the repair necessary as determined at inspection. On Post-SB1441 installation, a maximum of four washers may be used as spacers between the top and bottom mounting ring assemblies (Ref. IPC).

A. Removal of Compressor Wash Ring (Ref. Fig. 201)

CAUTION: KEEP INLET AREA CLEAR OF ALL FOREIGN OBJECTS WHICH COULD ENTER THE ENGINE AND CAUSE INTERNAL DAMAGE IF COMPRESSOR IS INADVERTENTLY TURNED.

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Figure 201 Removal/Installation of Compressor Wash Ring



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1. Spray Tube (RH)
2. Cover
3. Tee
4. Bolt
5. Washer
6. Self-locking Nut
7. Support Bracket
8. Top Fireseal Mount Ring
9. Spray Tube (LH)
10. Bolts
11. Self-locking Nut

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(1) Remove air inlet screen (Ref. 72-20-00).

(2) Loosen coupling nut and remove right-hand spray tube (1) from tee (3).

(3) Remove cover (2) and remove tee (3) from left-hand spray tube (9) by loosening coupling nut of tube.

(4) Remove left-hand spray tube (9) from front face of rear mount ring by removing two self-locking nuts (8), bolts (4) and washers (6). Exercise care as washers are fitted between mount ring and tube locating bracket (5).

B. Installation of Compressor Wash Ring (Ref. Fig. 201)

(1) Locate left-hand spray tube (9) on front face of rear fireseal and align mounting holes in tube locating bracket (5) with holes in fireseal mount ring and support bracket (7). Secure assembly with two bolts (4) (heads to be on front face of mount ring), washers (6) (between bracket (5) and mount ring) and self-locking nuts (8). Tighten nuts and torque 35 to 40 lb.in.

(2) Install tee (3) on tube (9). Coupling nut to be fingertight.

(3) Locate right-hand spray tube (1) on front face of fireseal mount ring and screw tube coupling nut onto tee (3), fingertight.

(4) Position tee at an angle of 20 degrees (± 5 degrees) relative to face of mount ring. Tighten coupling nuts of both spray tubes, torque 270 to 300 lb.in., and lockwire.

(5) Install cover (2) on tee and lockwire.

(6) Reinstall air inlet screen (Ref. 72-20-00).

6. Cleaning/Painting

A. Compressor Wash Ring (When fitted)

(1) Clean the tee and plugs with solvent (PWC11-027) or (PWC11-031). Dry with clean, dry, compressed air.

NOTE: If necessary, pressure flush the tee to make sure all the ports are clean and clear.

(2) Clean the spray tubes:

(a) Clean the exterior of the spray tubes with a soft cloth moistened in solvent (PWC11-027) or (PWC11-031).

(b) Pressure flush each spray tube:

1 Using a suitable flexible, coiled nonmetallic brush and a liberal quantity of petroleum solvent, thoroughly scrub the interior of each tube.

2 Temporarily tape over the spray holes in each tube.

3 Install a rubber hose over the plug end of each tube, and provide a suitable container for spillage.

4 Pressure flush through each tube from the coupling end, using petroleum solvent, and remove the rubber tubing.

5 Remove the tape from each tube, making sure that the tubes are completely free of

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tape residue.

6 Pressure flush with petroleum solvent, making sure all spray holes in each tube are clean and clear.

7 Remove the slave plugs and dry each tube with clean, dry, compressed air.

7. Inspection/Check

A. Rear Fireseal Mount Rings

(1) Examine the rear fireseal mount ring halves for attachment, damage and condition.

NOTE: For the external tubes/lines passing through the mount rings, refer to the relevant chapters in this manual.

(2) Examine the circumferential insulating strips for attachment. Loosened strips may be rebonded.

(3) Inspect mount ring spacers (Post-SB1441) for damage and condition. Replace as necessary.

B. Compressor Wash Ring

(1) Examine the tee and cap for condition/damage. Repair damaged threads with a suitable chaser/die.

(2) Examine the plugs for condition/damage. Replace the plugs as required.

(3) Inspect the spray tubes as follows:

(a) Check the spray holes in each tube. Clear the holes of any foreign matter with a 0.052 to 0.057 in. drill bit. DO NOT break sharp edges.

(b) Examine the brazed joints in the area of the coupling nuts and plugs. Cracks are not acceptable.

(c) Examine the locating brackets and support clips for proper attachment. Cracks are not acceptable.

(d) Dents which could alter the direction of the spray are not acceptable in the area of the holes.

(e) Inspect the tubes for general condition and surface damage as follows:

1 Scratches with no appreciable depth are acceptable. Scratches to 0.005 in. deep should be smoothed out. Replace the tube assembly if scratches exceed 0.005 in.

2 Nicks 0.0625 in. long by 0.010 in. wide and 0.003 in. deep are acceptable. Nicks to 0.005 in. should be smoothed out to remove sharp edges.

3 Dents are acceptable provided the length or depth does not exceed 10 percent of the tube OD nor greater than 5 percent in the swaged areas. No more than one dent, to the maximum depth, per 12-inch length of tubing is acceptable. Dents on tube bends, which can cause restriction by flattening and local weakening, are not acceptable.

4 Minor isolated pitting to 0.003 in. deep is acceptable. Clusters of pits not exceeding a maximum depth of 0.005 in should be blended.

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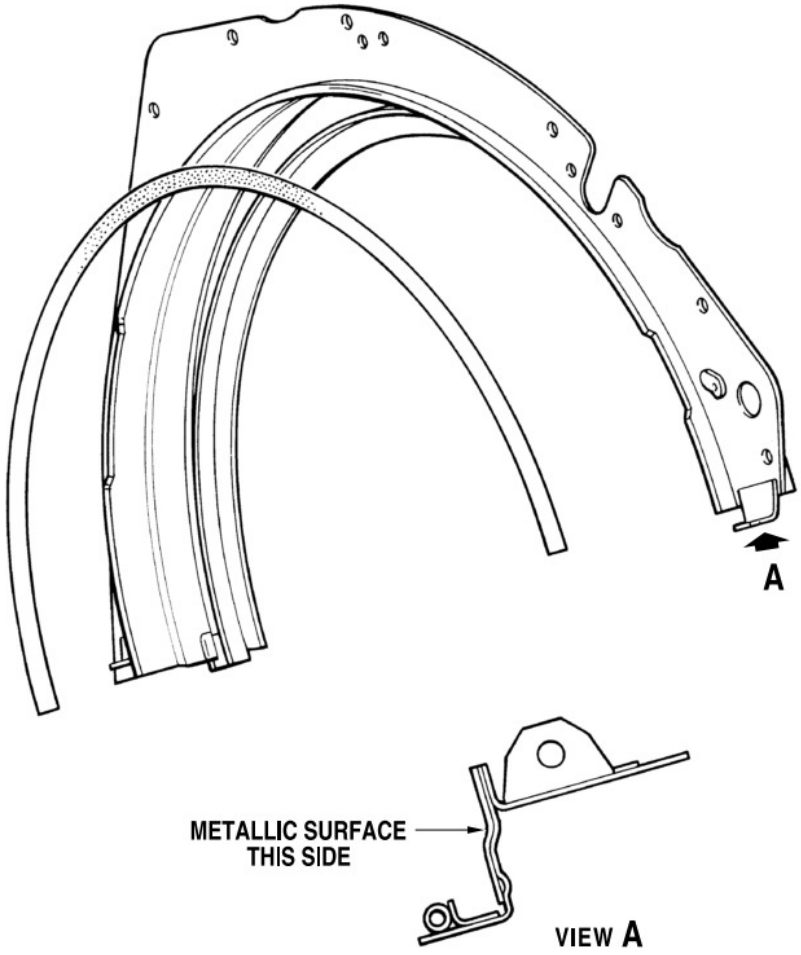
5 Corrosion, rust or stains are acceptable if they can be removed by light polishing with abrasive cloth (PWC05-061) and oil (PWC03-001).

8. Approved Repairs

A. Repair Mount Ring Insulating Strips (Ref. Fig. 202)

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Figure 202 Rear Fireseal Mount Ring - Insulating Strip Replacement



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(1) Detached fireseal insulating strips, if serviceable, may be rebonded:

NOTE: A new strip must be trimmed to the same length as that removed from fireseal.

(a) Remove the old adhesive from the mount ring and insulating strip with solvent (PWC11-027) or (PWC11-031).

(b) Bond the insulating strip to the mount ring halves using an adhesive sealant (PWC08-002).

(c) Allow the sealant to cure at room temperature for approximately 72 hours.

NOTE: Seventy two hours is the normal curing time of bond. The ring half may be returned to service after two to four hours, but avoid exerting pressure that would cause the sealant to exude and weaken bond.

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